

Product Profitability Review Policy

Internal analytical reference — synthetic policy for root-cause investigation

Products requiring investigation

- Products with persistently negative aggregate Profit should be investigated at product, country, customer and discount level.
- Products with high Sales but negative Profit should receive priority analytical attention because high sales volume can conceal poor economics.
- Where a product has only a small number of orders, the agent should state the limited sample size before making broader conclusions.

Category-level review triggers

- Tables, Machines and Bookcases should receive additional review when their aggregate ROS is negative.
- Supplies should also be investigated when negative profit persists despite relatively modest average discounts, because discount alone may not explain the outcome.
- High-ROS categories should not be assumed to be uniformly healthy; the agent should check whether loss-making products are being masked by profitable products within the category.

Diagnostic sequence

- Start with category → subcategory → product → country → customer → order.
- Compare Sales, Profit, ROS, Quantity and discount exposure at each level.
- Identify whether losses are broad-based or concentrated in a small number of products/customers/orders.
- Use the policy documents to generate hypotheses, then return to SQL/Python for numerical verification.

Observed-data rationale

- In the supplied data, Tables have the largest aggregate negative profit among the identified subcategories, while Machines and Bookcases are also loss-making.
- Several individual product-country combinations show concentrated losses and elevated discounts. These observations motivate the review triggers but do not establish the cause of the losses.